Measures of Center & Spread

Section 1: Mean, Median, and Skewness

1 In a strongly right-skewed distribution of household incomes, which relationship between the mean and median is typically expected?

a. The mean is greater than the median

b. The mean is less than the median

c. The mean equals the median

d. The relationship cannot be determined without sample size

2 Which summary statistic is considered resistant to extreme outliers?

a. Median and Interquartile Range (IQR)

b. Mean and Standard Deviation

c. Range and Mean

d. Standard Deviation and Variance

3 A dataset has values: $3, 5, 7, 8, 9, 11, 15$. If the maximum value 15 is replaced with 50, which statistic will NOT change?

a. Median

b. Mean

c. Range

d. Standard Deviation

Section 2: The 1.5 x IQR Rule for Outliers

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4 A dataset has $Q_1 = 24$ and $Q_3 = 40$. According to the $1.5 \times \text{IQR}$ rule, what are the lower and upper outlier boundaries?

a. Lower: 0; Upper: 64

b. Lower: 8; Upper: 56

c. Lower: 16; Upper: 48

d. Lower: 12; Upper: 60

5 If a distribution has five-number summary: $\text{Min} = 10$, $Q_1 = 20$, $\text{Med} = 28$, $Q_3 = 35$, $\text{Max} = 62$, which value is an outlier?

a. 62

b. 10

c. Both 10 and 62

d. Neither

6 What percentage of observations in any distribution lies strictly between Q_1 and Q_3?

a. Approximately 50%

b. 25%

c. 75%

d. 68%

Section 3: Standard Deviation and Variance

7 What does sample standard deviation s measure in a dataset?

a. The typical distance of data points from the sample mean

b. The distance between the maximum and minimum observations

c. The difference between the mean and median

d. The percentage of values that are negative

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8 If every observation in a sample has the exact same value of 14, what is the sample standard deviation?

a. 0

b. 14

c. 1

d. Undefined

9 Why is $(n-1)$ used in the denominator of the sample variance $s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$ instead of n?

a. To make s^2 an unbiased estimator of the population variance σ^2

b. To ensure the standard deviation is always an integer

c. Because one observation is always assumed to be an outlier

d. To adjust for potential measurement errors in the sample

Section 4: Comparing Distributions with Boxplots

10 A boxplot shows a long right whisker, a right-half box wider than the left-half box, and no outliers. What is the shape of the distribution?

a. Skewed to the right

b. Skewed to the left

c. Roughly symmetric

d. Uniform

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11 When comparing two test score distributions, School A has $\text{median} = 78, \text{IQR} = 12$ and School B has $\text{median} = 85, \text{IQR} = 6$. Which statement is justified?

a. School B typically scored higher and exhibited less variability in the middle 50% of scores

b. School A had a higher maximum score than School B

c. School A had higher consistency in student scores

d. School B had more outliers than School A

12 Which display allows you to recover the original raw data values?

a. Stem-and-leaf plot

b. Boxplot

c. Histogram

d. Cumulative relative frequency plot

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